Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

A horizontal spring block system of mass M executes simple harmonic motion. When the block is passing through its equilibrium position, an object of mass m is put on it and the two move together. Find the new amplitude

(A) $\sqrt{\frac{\mathrm{M}}{\mathrm{M+m}}}\mathrm{A}$

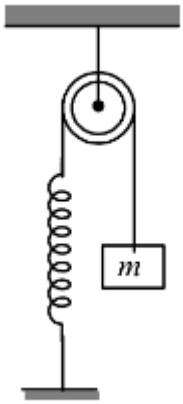
(B) $\sqrt{\frac{\mathrm{M+m}}{\mathrm{M}}}\mathrm{A}$

(C) $\sqrt{\frac{\mathrm{m}}{\mathrm{M+m}}}\mathrm{A}$

(D) $\sqrt{\frac{\mathrm{M+m}}{\mathrm{m}}}\mathrm{A}$

Q.2

A mass $m = 8\ \mathrm{kg}$ is attached to a spring as shown in figure and held in position so that the spring remains unstretched. The spring constant is $200\ \mathrm{N/m}$. The mass m is then released and begins to undergo small oscillations. The maximum velocity of the mass will be $\left(\mathrm{g} = 10\ \mathrm{m/s^2}\right)$



(A) $1\ \mathrm{m/s}$

(B) $2\ \mathrm{m/s}$

(C) $4\ \mathrm{m/s}$

(D) $5\ \mathrm{m/s}$

Q.3

The time period of a bar pendulum when suspended at distances $30\ \mathrm{cm}$ and $50\ \mathrm{cm}$ from its centre of gravity comes out to be the same if the mass of the body is $2\ \mathrm{kg}$ then its moment of inertia about an axis passing through first point will be :

(A) $4800\ \mathrm{kg-m^2}$

(B) $0.30\ \mathrm{kg-m^2}$

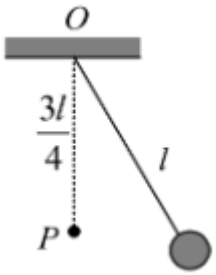
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(C) $0.48\ \text{kg} - \text{m}^2$

(D) $3000\ \text{kg} - \text{m}^2$

Q.4

A pendulum has time period T for small oscillations. An obstacle P $\frac{3l}{4}$. The is situated below the point of suspension O at a distance 4 . The pendulum is released from rest. Throughout the motion the moving string makes small angle with vertical. Time after which the pendulum returns back to its initial position is



(A) T

(B) $\frac{3T}{4}$

(C) $\frac{3T}{5}$

(D) $\frac{4T}{5}$

Q.5

Two linear S.H.M., having equal amplitude frequencies ω & 2ω are superimposed on a particle in x & y directions. If initial phase difference is $\pi/2$ between two S.H.M. then resultant path of particle is-

(A) $y = 4x^2\left(1 - \frac{x^2}{a^2}\right)$

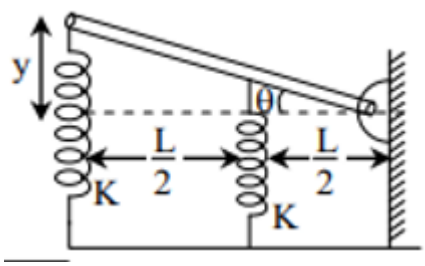
(B) $y^2 = x^2\left(1 - \frac{x^2}{a^2}\right)$

(C) $y^2 = 4x\left(1 - \frac{x^2}{a^2}\right)$

(D) $y^2 = 4x^2\left(1 - \frac{x^2}{a^2}\right)$

Q.6

In a gravity free space a long uniform rod of length L, mass M is free to rotate in a horizontal plane about a vertical axis through its end. Two springs of constant K each are connected as shown. On equilibrium, the rod was horizontal. The frequency of small oscillations in Hz will be -



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(A) $\frac{1}{2\pi}\sqrt{\frac{15\ \mathrm{K}}{\mathrm{M}}}$

(B) $\frac{1}{2\pi}\sqrt{\frac{5K}{4M}}$

(C) $\frac{1}{2\pi}\sqrt{\frac{3\ \mathrm{K}}{4\mathrm{M}}}$

(D) $\frac{1}{2\pi}\sqrt{\frac{15\ \mathrm{K}}{4\mathrm{M}}}$

Q.7

One solid sphere, having mass $1\ \mathrm{kg}$ & diameter $0.3\ \mathrm{m}$ is suspended from a wire. If the twisting couple per unit twist for the wire is $6 \times 10^{-3}\ \mathrm{N-m/radian}$, then the time period of small oscillations will be-

(A) $0.7\mathrm{sec}$

(B) $7.7\mathrm{sec}$

(C) $77\mathrm{sec}$

(D) $777\mathrm{sec}$

Q.8

A particle of mass m is located at a place where it is acted upon by a one dimensional conservative force and its potential energy is given by: $U(x) = A(1 - \cos px)$, where A and p are constants. The period of small oscillations of the particle is:

(A) $2\pi\sqrt{\frac{m}{(Ap)}}$

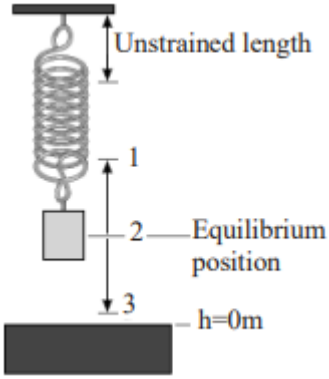
(B) $2\pi\sqrt{\frac{m}{(Ap^2)}}$

(C) $2\pi\sqrt{\frac{m}{A}}$

(D) $\frac{1}{2\pi}\sqrt{\frac{Ap}{m}}$

Q.9

The figure shows an object attached to an ideal spring, which is hanging from the ceiling. The unstrained length of the spring is indicated. For purposes of measuring the height h that determines the gravitational potential energy, the floor is taken as the position where $\mathrm{h} = 0\ \mathrm{m}$. The equilibrium position at which the object hangs stationary is identified as position 2. The object is set into vertical simple harmonic motion between positions 1 and 3. Choose the incorrect statements -



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(A) KE has a maximum value at position 2. EPE has a maximum value at position 1. GPE has a maximum value at position 3 .

(B) KE has a maximum value at position 1. EPE has a maximum value at position 3. GPE has a maximum value at position 2 .

(C) KE has a maximum value at position 3. EPE has a maximum value at position 2. GPE has a maximum value at position 1 .

(D) KE has a maximum value at position 2. EPE has a maximum value at position 3. GPE has a maximum value at position 1.

Q.10

Two particles P and Q are performing SHM about same mean position with same time period ' T but their amplitude are $\mathrm{A} \& 2\ \mathrm{A}$ respectively. At $\mathrm{t}=0, \mathrm{P}$ is at $+\frac{\mathrm{A}}{2}, \mathrm{Q}$ is at $2\ \mathrm{A}$ and both are moving towards their mean position. Choose the CORRECT statement(s) :-

(A) Phase difference between them remains constant.

(B) Initial phase difference between P and Q is $\frac{2\pi}{3}$.

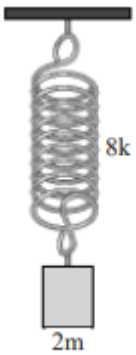
(C) They will cross each other after time $\frac{T}{3}$.

(D) Maximum separation between P & Q during the motion will be $\sqrt{3}\ \mathrm{A}$.

Q.11

An external force (in addition to the spring force) is continually applied to an object of mass m attached to a spring that has a spring constant k. The frequency of this external force is such that resonance occurs. Then the frequency of this external force is doubled. In which of the following figures resonance will not occur.

(A)

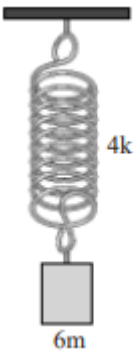


(B)



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(C)



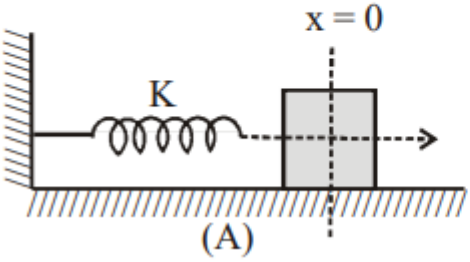
(D)

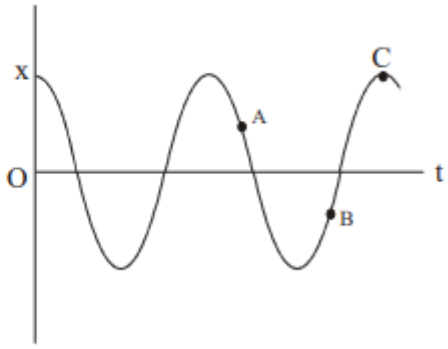


Q.12

Passage:

Block of mass 100gm is attached with the spring of force constant $25.6\ \text{N/m}$ as shown in figure. Block is free to oscillate on frictionless horizontal surface. Block start performing SHM from $x = 0$ at $\text{t} = 0$ in positive x-direction. Position-time graph of block is shown in figure.





Question: In curve when block is at position $\text{A}-$

(A) Its velocity and displacement both negative.

(B) Its velocity is positive and displacement is negative

(C) Its velocity is negative and displacement is positive

(D) Its velocity and displacement both positive.

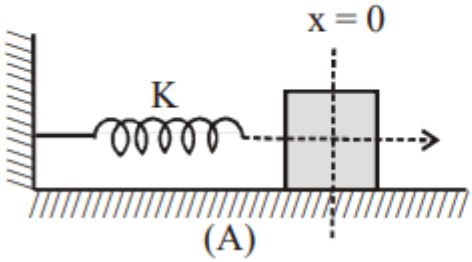
Q.13

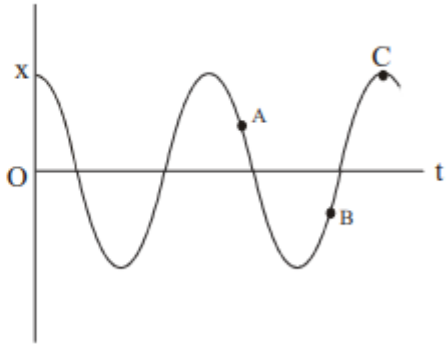
Passage:

Block of mass 100gm is attached with the spring of force constant $25.6\ \text{N/m}$ as shown in figure. Block is free to oscillate on frictionless horizontal surface. Block start performing SHM from $x = 0$ at $\text{t} = 0$ in positive x-

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direction. Position-time graph of block is shown in figure.





Question: In curve when block is at position B -

(A) Its velocity and displacement both negative.

(B) Its velocity is positive and displacement is negative

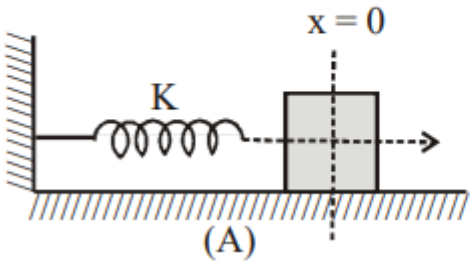
(C) Its velocity is negative and displacement is positive.

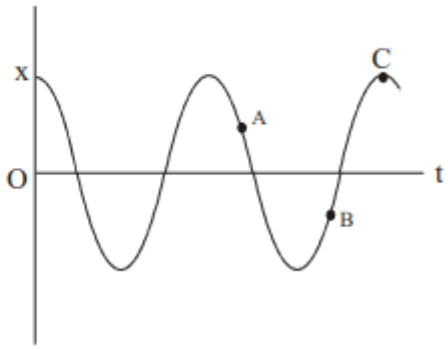
(D) Its velocity and displacement both positive.

Q.14

Passage:

Block of mass 100gm is attached with the spring of force constant $25.6\ \text{N/m}$ as shown in figure. Block is free to oscillate on frictionless horizontal surface. Block start performing SHM from $x = 0$ at $\text{t} = 0$ in positive x-direction. Position-time graph of block is shown in figure.





Question: In curve when block is at position $\text{C}-$

(A) Velocity is maximum and acceleration is zero.

(B) Velocity is minimum and acceleration is zero.

(C) Velocity is zero and acceleration is negative.

(D) Velocity is zero and acceleration is positive.

Q.15

Passage:

A $2\ \text{kg}$ block hangs without vibrating at the bottom end of a spring with a force constant of $800\ \text{N/m}$. The top end of the spring is attached to the ceiling of an elevator car. The car is rising with an upwards acceleration of $10\ \text{m/s}^2$ when the acceleration suddenly ceases at time $\text{t} = 0$ and the car moves upward with constant speed.

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$\left(g = 10\ m/s^2\right)$

Question: What is the angular frequency of oscillation of the block after the acceleration ceases ?

(A) $10\sqrt{2}rad/s$

(B) $20rad/s$

(C) $20\sqrt{2}rad/s$

(D) $32rad/s$

Q.16

Passage:

A $2\ kg$ block hangs without vibrating at the bottom end of a spring with a force constant of $800\ N/m$. The top end of the spring is attached to the ceiling of an elevator car. The car is rising with an upwards acceleration of $10\ m/s^2$ when the acceleration suddenly ceases at time $t = 0$ and the car moves upward with constant speed.

$\left(g = 10\ m/s^2\right)$

Question: The amplitude of the oscillation is -

(A) $7.5\ cm.$

(B) $5\ cm.$

(C) $2.5\ cm.$

(D) $1\ cm.$

Q.17

Passage:

A $2\ kg$ block hangs without vibrating at the bottom end of a spring with a force constant of $800\ N/m$. The top end of the spring is attached to the ceiling of an elevator car. The car is rising with an upwards acceleration of $10\ m/s^2$ when the acceleration suddenly ceases at time $t = 0$ and the car moves upward with constant speed.

$\left(g = 10\ m/s^2\right)$

Question: The initial phase angle observed by a rider in the elevator, taking downward direction to be positive and positive extreme position to have $\pi/2$ phase constant, is equal to -

(A) zero

(B) $\pi/2rad$

(C) πrad

(D) $3\pi/2rad$

Q.18

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Match the column -

Column I	Column II
(A) Periodic	(p) $y = \sin kt + \cos kt$
(B) Periodic and SHM	(q) $y = \cos(kt) + 2\sin^2(kt)$
(C) Periodic and not SHM	(r) $y = e^{-kt}$
(D) Not periodic and not SHM	(s) $y = \tan(kt)$

(A) A-(pqs) B-(r) C-(qs) D-(r)

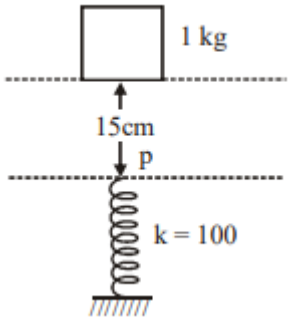
(B) A-(pqs) B-(p) C-(qsr) D-(p)

(C) A-(pqs) B-(p) C-(qs) D-(r)

(D) A-(pqs) B-(qr) C-(s) D-(sp)

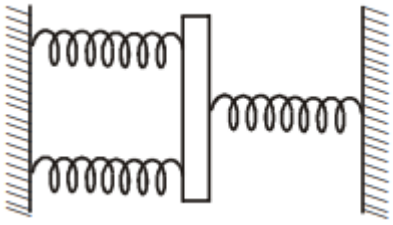
Q.19

In the set up shown, $1\ \text{kg}$ mass is positioned 15cms above the end P of a string in vertical position, its spring constant is $100\ \text{N/m}$. The mass is dropped and reaches the end P of the spring at time t. After $0.1\pi + \frac{\pi}{10\lambda}$ (second) time delay, the mass come back to the same position. Find the value of λ.



Q.20

The three springs are identical. The bar of mass m performs SHM on horizontal plane such that its kinetic energy T varies at n cycles/s. Then the spring constant of each spring is $\frac{\text{m}\pi^2\text{n}^2}{\text{A}}$. Find the value of A.



Answers & Solutions

Q.1 Answer:

—

Solution:

Suppose the velocities in the equilibrium position before and after placing the object m and v and v', and the amplitudes are A and A' respectively.

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By conservation of energy, we have $\frac{1}{2}\mathrm{Mv}^2 = \frac{1}{2}\mathrm{KA}^2 ...(1)$

and $\frac{1}{2}(\mathrm{M}+\mathrm{m})\mathrm{v}'^2 = \frac{1}{2}\ \mathrm{K}\ \mathrm{A}'^2 ...(2)$

and by conservation of momentum $\mathrm{Mv} = (\mathrm{M}+\mathrm{m})\mathrm{v}'$

Hence $\frac{1}{2}\frac{\mathrm{M}^2\mathrm{v}^2}{(\mathrm{M}+\mathrm{m})} = \frac{1}{2}\mathrm{KA}'^2 ...(3)$

Dividing (3) by (1) we get $\frac{\mathrm{M}}{(\mathrm{M}+\mathrm{m})} = \frac{\mathrm{A}'^2}{\mathrm{A}^2} \Rightarrow \mathrm{A}' = \sqrt{\frac{\mathrm{M}}{\mathrm{M}+\mathrm{m}}}\mathrm{A}$

Q.2 Answer:

$2\ \mathrm{m/s}$

Solution:

Mean position will be at $k \quad x = mg \quad$ **or** $x = \frac{mg}{k} = \frac{8\times10}{200} = \frac{2}{5}m = 0.4\ \mathrm{m}$

i.e. $A = 0.4\ \mathrm{m}$

Now, $v_{\max} = A\omega = A\sqrt{\frac{k}{m}} = (0.4)\sqrt{\frac{200}{8}} = 2\ \mathrm{m/s}$

$\therefore \quad$ **(b)**

Q.3 Answer:

$0.48\ \mathrm{kg}-\mathrm{m}^2$

Solution:

For point -1

$\tau = \mathrm{mg}\left(\frac{30}{100}\right)\sin\Theta$

As Θ **is very small**

$$\tau = \frac{3\mathrm{mg}}{10}\Theta$$

$$\mathrm{C}_1 = \frac{3\mathrm{mg}}{10}$$

where t point 2

$$\tau = \frac{5}{10}\mathrm{mg}\Theta$$

$$\mathrm{C}_2 = \frac{5}{10}\mathrm{mg}$$

As the period of $\mathrm{SHM}(T) = \Theta n\sqrt{\frac{I}{C}}$

Let at COM I is the moment of inertia

By parallel axis theorem

$$\mathrm{I}_1 = \mathrm{I} + (2)\left(\frac{3}{10}\right)^2$$

$$\mathrm{I}_2 = \mathrm{I} + 2\left(\frac{5}{10}\right)^2$$

Because time period of blue points are equal

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$$2\pi\sqrt{\frac{I_1}{c_1}} = 2\pi\sqrt{\frac{T_2}{c_2}} \Rightarrow \frac{I_1}{I_2} = \frac{c_1}{c_2}$$

$$\Rightarrow \frac{I + 2\left(\frac{3}{10}^2\right)}{I + 2\left(\frac{5}{10}\right)^2} = \frac{\frac{3mg}{10}}{\frac{5mg}{10}} = \frac{3}{5}$$

$$\Rightarrow I = 0.30 \text{ kg m}^2$$

At point $-1, I_1 = 0.30 + 2\left(\frac{3}{10}\right)^2 = 0.48 \text{ kg m}^2$

Q.4 Answer:

$\frac{3T}{4}$

Solution:

After P length of pendulum becomes $\frac{l}{4}$.
Now, as $T \propto \sqrt{l}$, so after P time period will become $T' = T/2$
Therefore, the desired time will be $t = \frac{T}{2} + \frac{T'}{2} = \frac{T}{2} + \frac{T}{4} = \frac{3T}{4}$
$\therefore$ **(b)**

Q.5 Answer:

$y^2 = 4x^2\left(1 - \frac{x^2}{a^2}\right)$

Solution:

Equation of motion in x direction, $x = a\cos\omega t$ Equation of motion in y direction, $y = a\cos(2\omega t + \pi/2)$

$y = -a\sin 2\omega t$

$y = -2a\sin\omega t\cos\omega t$

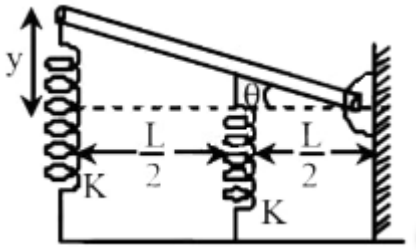
On eliminating t, $y = -2a\sqrt{1 - \frac{x^2}{a^2}}\left(\frac{x}{a}\right); y^2 = 4x^2\left(1 - \frac{x^2}{a^2}\right)$

Q.6 Answer:

$\frac{1}{2\pi}\sqrt{\frac{15\text{ K}}{4M}}$

Solution:

Displacement of the spring be y and y_1



As the image is given we can calculate the value of θ is, $\theta = \frac{y}{L}$
Also, $\theta = \frac{y_1}{\frac{L}{2}}$
$\therefore y = L\theta$ and $y_1 = \frac{L\theta}{2}$
Now let the force applied on the two rods are F_1 and F_2

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Hence the spring constant is K

$$F_1 = Ky = KL\theta$$

$$F_2 = Ky_1 = \frac{KL\theta}{2}$$

The restoring Torque due to the elastic force of the spring is:

$$\tau = \left(F_1 \cos\theta \frac{L}{2} \times F_2 \cos\theta L\right)$$

$$\Rightarrow \tau = \left(\frac{F_1}{2} + F_2\right)\cos\theta$$

Here θ is very small, hence we can assume $\cos\theta \simeq 1$

$$\therefore \tau = \left(\frac{KL\theta}{4} + KL\theta\right)L$$

$$= \left(\frac{K\theta}{4} + K\theta\right)L^2$$

$$= \left(\frac{5KL\theta}{4}\right)L^2$$

Now, Net Torque $\tau = \left(\frac{5KL\theta}{4}\right)L^2 = -I\alpha (\alpha =$ angular acceleration $)$

$$\Rightarrow \left(\frac{5KL\theta}{4}\right)L^2 = \left(\frac{ML^2}{3}\right)\alpha \left(\because I = -\left(\frac{ML^2}{3}\right)\right)$$

$$\Rightarrow \alpha = -\frac{15K}{4m}\theta$$

As we know $\alpha = -\omega^2\theta$

Putting the value we get, $\omega^2 = \frac{15K}{4m}$

$\because \omega = 2\pi f = \sqrt{\frac{15K}{4m}}$

Hence, the value of frequency $f = \frac{1}{2\pi}\sqrt{\frac{15K}{4m}}$

Q.7 Answer:

7.7sec

Solution:

$\mathrm{T} = 2\pi\sqrt{\frac{\mathrm{I}}{\mathrm{C}}} \ldots (1)$

$\mathrm{I} = \frac{2}{5}\mathrm{MR}^2 \ldots (2)$

By equation (1) & (2) $\mathrm{T} = 2\pi\sqrt{\frac{2\mathrm{MR}^2}{5\mathrm{C}}}$

Given, $\mathrm{M} = 1\ \mathrm{kg}, \mathrm{R} = 0.15\ \mathrm{m}, \mathrm{C} = 6 \times 10^{-3}\ \mathrm{N-m/rad.m}, \mathrm{T} = ?$

$\Rightarrow \mathrm{T} = 6.28\sqrt{\frac{0.4 \times 1 \times (0.15)^2}{6 \times 10^{-3}}};$

$\mathrm{T} = 6.28\sqrt{\frac{9 \times 10^{-3}}{6 \times 10^{-3}}} = 6.28 \times \sqrt{1.5};$

$\mathrm{T} = 7.7\text{sec}$

Q.8 Answer:

$2\pi\sqrt{\frac{m}{(Ap^2)}}$

Solution:

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Here, $V(x) = A(1 - \cos px)$

Force, $F = \frac{dv}{dx} = -\frac{d}{dx}(A - A\cos px)$

For small x, $F = -Ap^2x$

Acceleration, $a = \frac{F}{m} = \frac{Ap^2x}{m}$

The standard equation of SHM is, $a = -\omega^2 x$

Comparing equations (i) and (ii), we get

(ii)

$\omega^2 = \frac{Ap^2}{m}$ or $\omega = \sqrt{\frac{Ap^2}{m}}$

Period of oscillation, $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{Ap^2}}$

Q.9 Answer:

KE has a maximum value at position 2. EPE has a maximum value at position 1. GPE has a maximum value at position 3 .

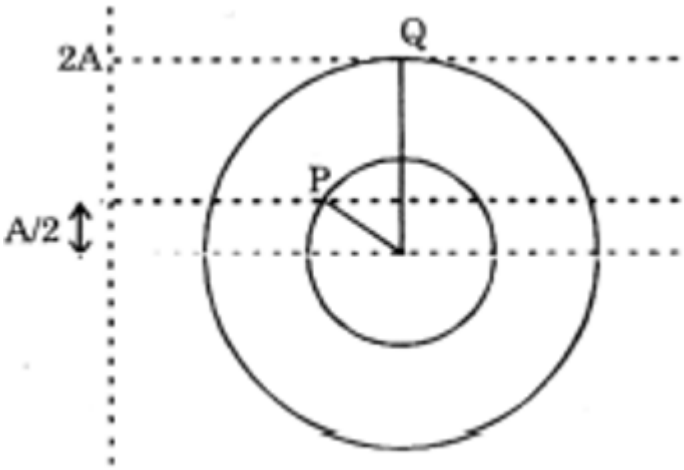
Solution:

In simple harmonic motion the speed and, hence, KE has a maximum value as the object passes through its equilibrium position, which is position 2. EPE has a maximum value when the spring is maximally stretched at position 3. GPE has a maximum value when the object is at its highest point above the ground, that is, at position 1**.**

Q.10 Answer:

Phase difference between them remains constant.

Solution:



Initial phase difference $= \frac{\pi}{3}$ **and since time period is same thus phase difference remains constant.**

Also,

$$X_P = A\sin\left(\omega t + \frac{5\pi}{6}\right)$$

$$\Rightarrow \mathrm{v}_P = A\omega\cos\left(\omega t + \frac{5\pi}{6}\right)$$

$$X_Q = 2A\sin\left(\omega t + \frac{\pi}{2}\right)$$

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Q.11 Answer:



Solution:

Resonance occurs when the frequency of the external force equals the frequency of oscillation of the object on the spring. The angular frequency of such a system is given by Equation $\left(\omega = \sqrt{\frac{\mathrm{k}}{\mathrm{m}}}\right)$.

Since the frequency of the force is doubled, the new frequency must be $2\omega = 2\sqrt{\frac{\mathrm{k}}{\mathrm{m}}}$.

The frequency of system A **is, in fact,** $\omega = \sqrt{\frac{8\mathrm{k}}{2\,\mathrm{m}}} = 2\sqrt{\frac{\mathrm{k}}{\mathrm{m}}}$.

Q.12 Answer:

Its velocity is negative and displacement is positive

Solution:

In curve when block is at position A then Its velocity is negative and displacement is positive.

Q.13 Answer:

Its velocity is positive and displacement is negative

Solution:

In curve when block is at position B then its velocity is positive and displacement is negative

Q.14 Answer:

Velocity is zero and acceleration is negative.

Solution:

In curve when block is at position C **then velocity is zero and acceleration is negative.**

Q.15 Answer:

$20\mathrm{rad/s}$

Solution:

$$\omega = \sqrt{\frac{\mathrm{k}}{\mathrm{m}}} = \sqrt{400} = 20\mathrm{rad/s}$$

Q.16 Answer:

$2.5\ \mathrm{cm}.$

Solution:

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$\omega = \sqrt{\frac{\mathrm{k}}{\mathrm{m}}} = \sqrt{400} = 20\mathrm{rad/s}$

Maximum extension the spring from natural position is x.

Then $mg + ma = kx \Rightarrow x = \frac{2(10+10)m}{800} = 5$ **cm.**

Extension of the spring when it is stretched to equilibrium line is x '$.\mathrm{mg} = \mathrm{kx}' \Rightarrow \mathrm{x}' = \frac{2\times10}{800} = 2.5$ cm

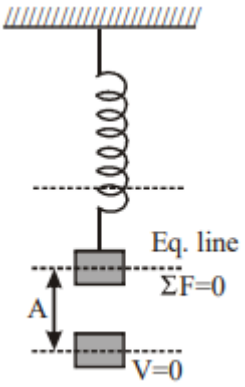
Q.17 Answer:

$\pi/2\mathrm{rad}$

Solution:

Using $x = A\sin(\omega t + \phi)$

$A = A\sin\phi \Rightarrow \phi = \pi/2$



Q.18 Answer:

A-(pqs) B-(p) C-(qs) D-(r)

Solution:

$(A) - pqs, (B) - p, (C) - qs, (D) - r$

For SHM, $\frac{\mathrm{d}^2\mathrm{y}}{\mathrm{dt}^2} \propto \mathrm{y}$

The periodic motion may or may not be oscillatory but SHM is always oscillatory.

Q.19 Answer:

—

Solution:

The movement of mass after it reaches P **from top can be analysed as shown. If** y **is the total compression it consists of two sections, i.e. distance** x **when it reach the neutral position of oscillation and distance A when it reach the maximum compression, A being the amplitude of oscillations,** $\mathrm{y} = \mathrm{A} + \mathrm{x}$.

Energy equation : $\mathrm{mgh} + \mathrm{mgy} = \frac{1}{2}\mathrm{ky}^2$

$1 \times 10 \times 0.15 + 1 \times 10 \times y = \frac{1}{2} \times 100y^2$

$50\mathrm{y}^2 - 10\mathrm{y} - 1.5 = 0 \Rightarrow \mathrm{y} = \frac{10\pm\sqrt{100+300}}{100} = \frac{10+20}{100} = 0.3$ m

At neutral position $\mathrm{Q} \Rightarrow \mathrm{mg} = \mathrm{kx}$

$\Rightarrow \quad x = \frac{mg}{k} = \frac{1\times10}{100} = 0.1$ m

$\therefore$ **Amplitude** $A = 0.3 - 1 = 0.2$ m

Now the time to reach $\mathrm{P} \rightarrow \mathrm{Q}$ **will be same as** $\mathrm{Q} \rightarrow \mathrm{P}$ **on return.**

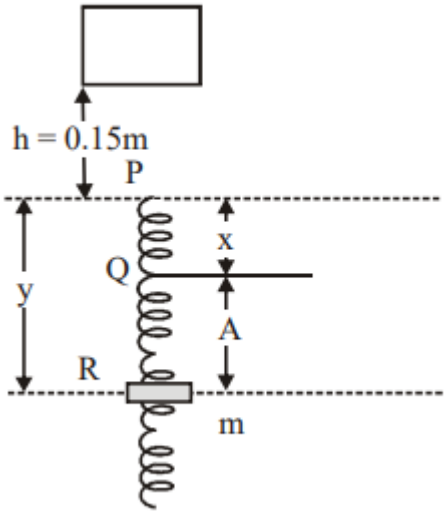
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$QP = x = A\sin\omega t$ **where** $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{100}{1}} = 10 rad/s$

$\sin 10t = \frac{x}{A} = \frac{0.1}{0.2} = \frac{1}{2} \Rightarrow 10t = \frac{\pi}{6} \Rightarrow t = \frac{\pi}{60}\ s \Rightarrow \omega = \frac{2\pi}{T} = 10 \Rightarrow T = \frac{2\pi}{10} = 0.2\pi$

Total time for the mass t **travel** P **to** R **and back** $= \frac{T}{2} + 2t = 0.1\pi + \frac{\pi}{30} = 0.42\ s$

$\Rightarrow 0.1\pi + \frac{\pi}{3\times 10} = 0.42$ **s. So,** $\lambda = 3$



Q.20 Answer:

—

Solution:

The arrangement is equivalent to three springs in parallel.

$\therefore$ **Equivalent spring constant** $= k' = 3k$

If angular frequency of displacement is ω' **then**

$\omega' = \frac{2\pi n}{2} = \pi n;$

$\omega'^2 = \frac{k'}{m} = \frac{3k}{m} \Rightarrow k = \frac{m\omega'^2}{3} = m\frac{\pi^2 n^2}{3}$. **So,** $A = 3$

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